

LKS SMK
Tingkat Nasional Ke-XXIV
Tahun 2016



LKS SMK

Tingkat Nasional XXIV
Malang, 23 – 27 Mei 2016

Soal

BIDANG LOMBA

IT-Software Application



KEMENTERIAN PENDIDIKAN DAN KEBUDAYAAN

DIREKTORAT JENDERAL PENDIDIKAN MENENGAH

DIREKTORAT PEMBINAAN SEKOLAH MENENGAH KEJURUAN

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PROJECT OVERVIEW

In this Test Project, you are required to develop an information system following the requirement given on the Test Project. In general, there will be 4 project resources given to you, which is:

1. ERD and Data Dictionary for Test Project

This will be used to guide you to create the database files. Ensure that all entities created on the database are related with the given ERD, following the relationship and also the criteria in Data Dictionary for each entity!

2. Navigation Diagram for the Application

Use the Navigation Diagram to develop the form interaction inside the application. You are free to add new interaction between forms, without neglecting the defined interaction in this diagram.

3. Example Design (Wireframe) of desired Information System

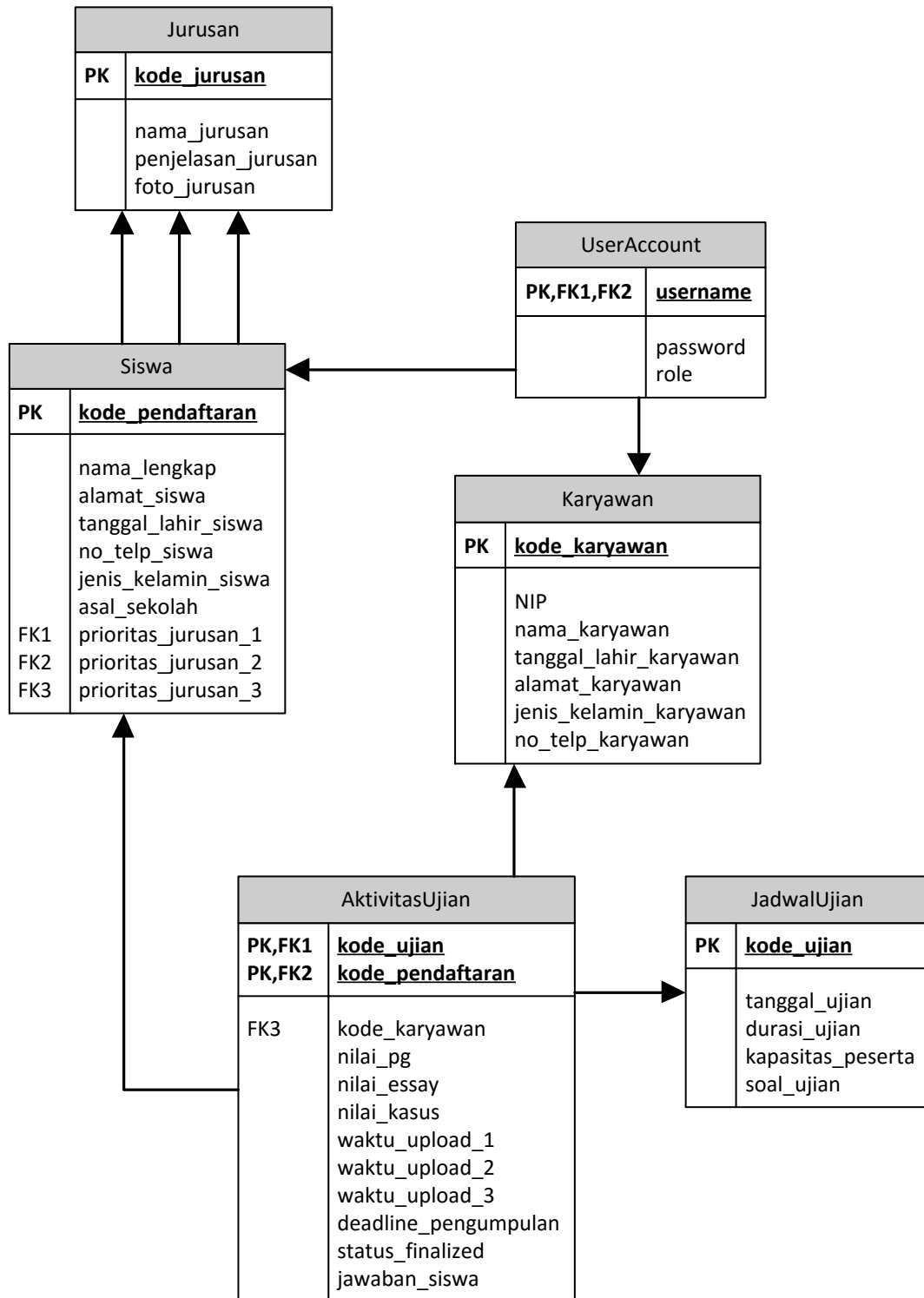
This file is used to give you design guidelines of all required forms. Please note that your form designs are not limited to these examples!

4. Data files (*if any*)

The data files will be used to support you test and develop the information system on each module. Please use this data files in the development, you are allowed to use another resources outside the given data files only if instructed in the module.

SMK NUSANTARA INFORMATION SYSTEM

ENTITY RELATIONSHIP DIAGRAM



SMK NUSANTARA INFORMATION SYSTEM

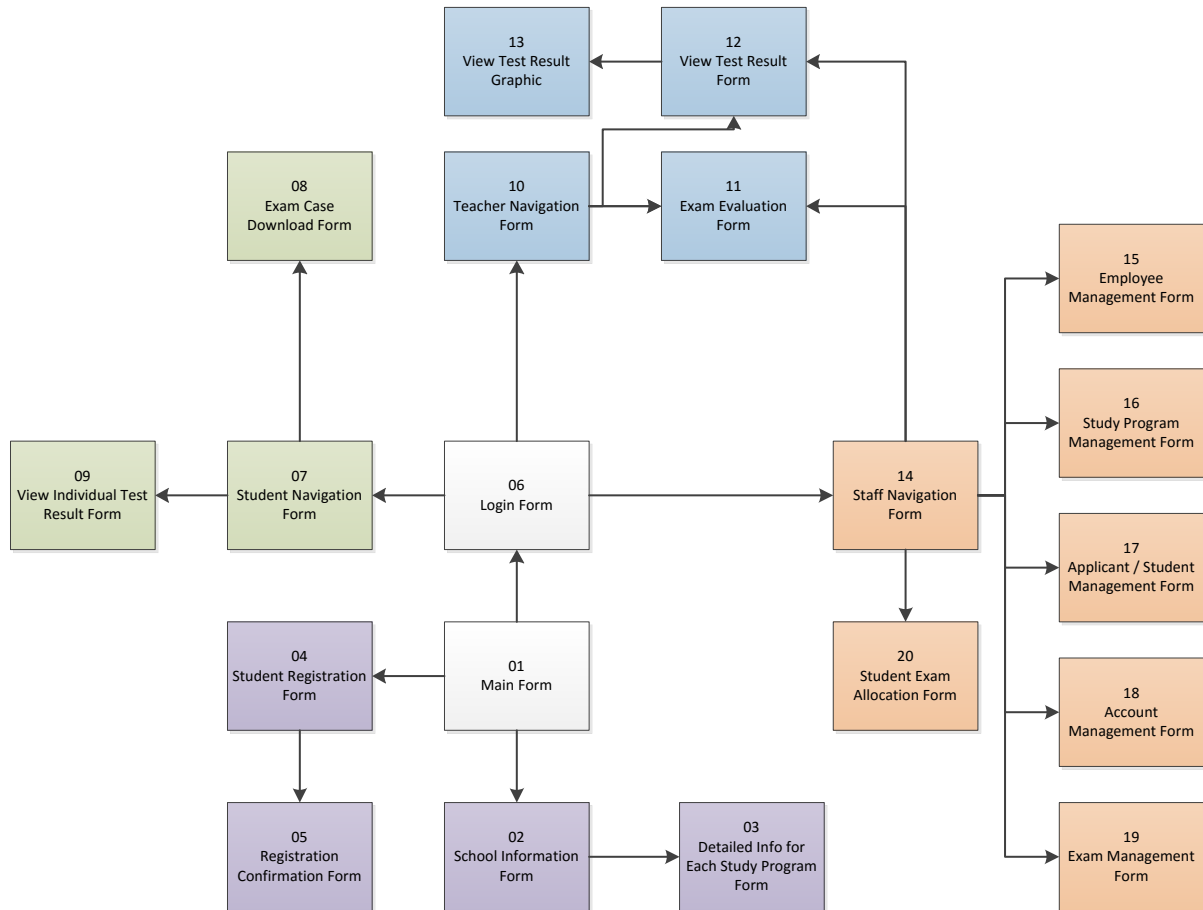
DATA DICTIONARY

Tabel	KEY	Kolom	Tipe Data	Required?	Notes
Jurusan	PK	kode_jurusan	char(5)	Y	Auto Increment
		nama_jurusan	varchar(100)	Y	
		penjelasan_jurusan	text	Y	
		foto_jurusan	text	Y	
UserAccount	PK, FK	username	char(5)	Y	
		password	text	Y	
		role	int	Y	
Siswa	PK	kode_pendaftaran	char(5)	Y	Auto Increment
		nama_lengkap	varchar(100)	Y	
		alamat_siswa	text	Y	
		tanggal_lahir_siswa	date	Y	
		no_telp_siswa	varchar(20)	Y	
		jenis_kelamin_siswa	varchar(20)	Y	
		asal_sekolah	text	Y	
	FK	prioritas_jurusan_1	char(5)	Y	
	FK	prioritas_jurusan_2	char(5)	Y	
	FK	prioritas_jurusan_3	char(5)	Y	
Karyawan	PK	kode_karyawan	char(5)	Y	Auto Increment
		NIP	text	Y	
		nama_karyawan	varchar(100)	Y	
		tanggal_lahir_karyawan	date	Y	
		alamat_karyawan	text	Y	
		jenis_kelamin_karyawan	varchar(20)	Y	
		no_telp_karyawan	varchar(20)	Y	
JadwalUjian	PK	kode_ujian	char(5)	Y	Auto Increment
		tanggal_ujian	date	Y	
		durasi_ujian	int	Y	
		kapasitas_peserta	int	Y	
		soal_ujian	varchar()	Y	

Tabel	KEY	Kolom	Tipe Data	Required?	Notes
AktivitasUjian	PK, FK	kode_ujian	char(5)	Y	
	PK, FK	kode_pendaftaran	char(5)	Y	
		kode_karyawan	char(5)	Y	
		nilai_pg	int		
		nilai_essay	int		
		nilai_kasus	int		
		waktu_upload_1	datetime		
		waktu_upload_2	datetime		
		waktu_upload_3	datetime		
		deadline_pengumpulan	datetime		
		status_finalized	varchar(20)		
		jawaban_siswa	text		

SMK NUSANTARA INFORMATION SYSTEM

NAVIGATION DIAGRAM



SMK NUSANTARA INFORMATION SYSTEM

WIREFRAME

01 Main Form

The wireframe shows a web application window titled "Pendaftaran SMK Nasional". The header area contains the school name "SMK Nasional" in large bold letters, the academic year "Tahun Ajaran 2016/2017" below it, and a "Login Pengguna" button on the right. The main content area features two large buttons: "Pendaftaran Siswa Baru" on the left and "Informasi Sekolah" on the right. The "Pendaftaran Siswa Baru" button is highlighted with a blue border.

02 School Information Form

The wireframe shows a web application window titled "Informasi Sekolah". The header area contains the school name "SMK Nasional" in large bold letters and a "Kembali" button on the right. The main content area is divided into two columns. The left column features a photograph of the school building. The right column contains three paragraphs of text describing the school's history, fields of study, and staff. Below the text, there is a section titled "Informasi Jurusan" with a dropdown menu currently showing "Rekayasa Piranti Lunak". At the bottom right, there is a "Lihat Informasi" button.

03 Detailed Info for Each Study Program Form

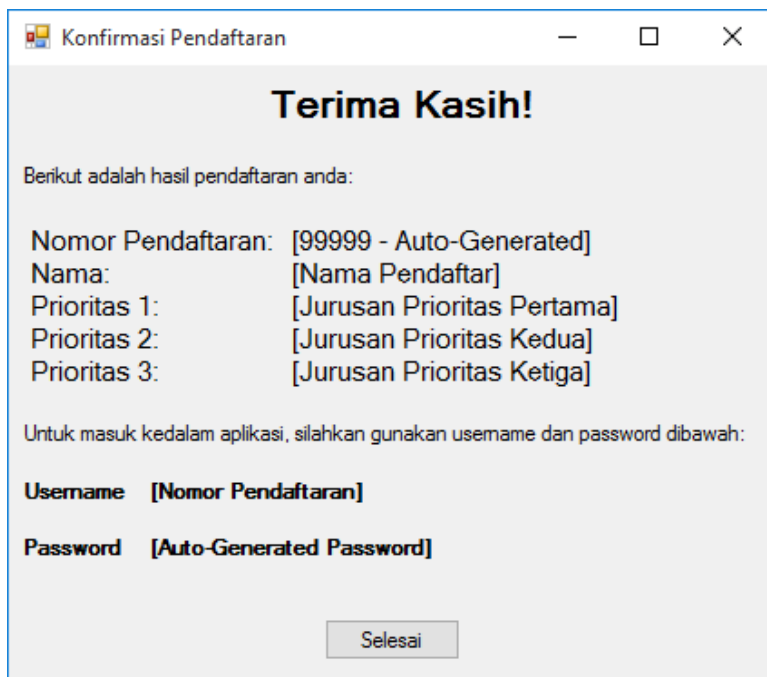
The screenshot shows a web browser window with the title "[Nama Jurusan]". In the top right corner, there is a "Kembali" button. The main heading is "[Nama Jurusan]". Below it is a large gray rectangular box labeled "Image". Underneath the image box is the text "[Penjelasan mengenai jurusan yang dipilih]". At the bottom, there is a paragraph of Lorem Ipsum placeholder text.

04 Student Registration Form

The screenshot shows a web browser window with the title "Pendaftaran Siswa Baru". In the top right corner, there is a "Kembali" button. The main heading is "Pendaftaran Siswa Baru". The form contains the following fields and elements:

- Nama Lengkap**: A text input field.
- Alamat**: A text input field.
- Tanggal Lahir**: A date picker showing "Friday, December 11, 2015".
- No. Telepon**: A text input field.
- Jenis Kelamin**: Two radio buttons labeled "Laki-Laki" and "Perempuan".
- Asal Sekolah**: A text input field.
- Pilihan Jurusan**: A section containing three dropdown menus labeled "Prioritas 1", "Prioritas 2", and "Prioritas 3".
- Masukkan kode berikut untuk melanjutkan pendaftaran:**: A text input field with the code "xYz1\$3mh9!" and a "DAFTAR" button.

05 Registration Confirmation Form



Konfirmasi Pendaftaran

Terima Kasih!

Berikut adalah hasil pendaftaran anda:

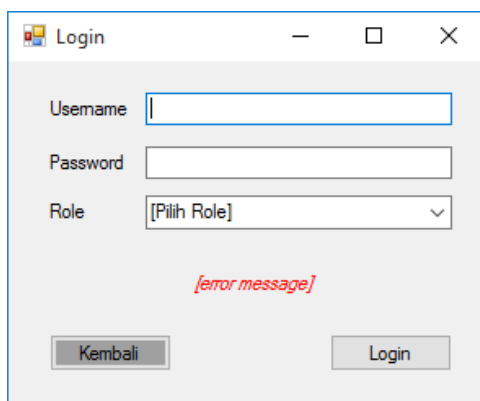
Nomor Pendaftaran: [99999 - Auto-Generated]
Nama: [Nama Pendaftar]
Prioritas 1: [Jurusan Prioritas Pertama]
Prioritas 2: [Jurusan Prioritas Kedua]
Prioritas 3: [Jurusan Prioritas Ketiga]

Untuk masuk kedalam aplikasi, silahkan gunakan username dan password dibawah:

Username [Nomor Pendaftaran]
Password [Auto-Generated Password]

Selesai

06 Login Form



Login

Username

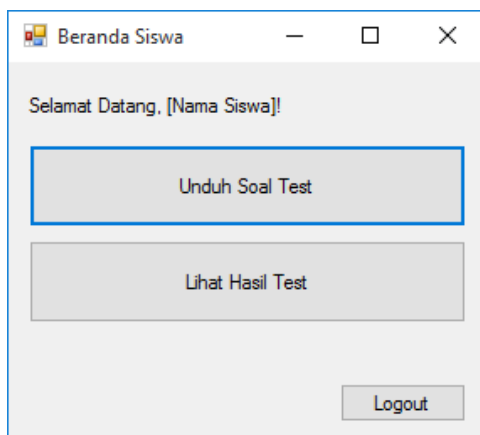
Password

Role

[error message]

Kembali Login

07 Student Navigation Form



Beranda Siswa

Selamat Datang, [Nama Siswa]!

Unduh Soal Test

Lihat Hasil Test

Logout

08 Exam Case Download Form

Download Soal Test

Jadwal Ujian

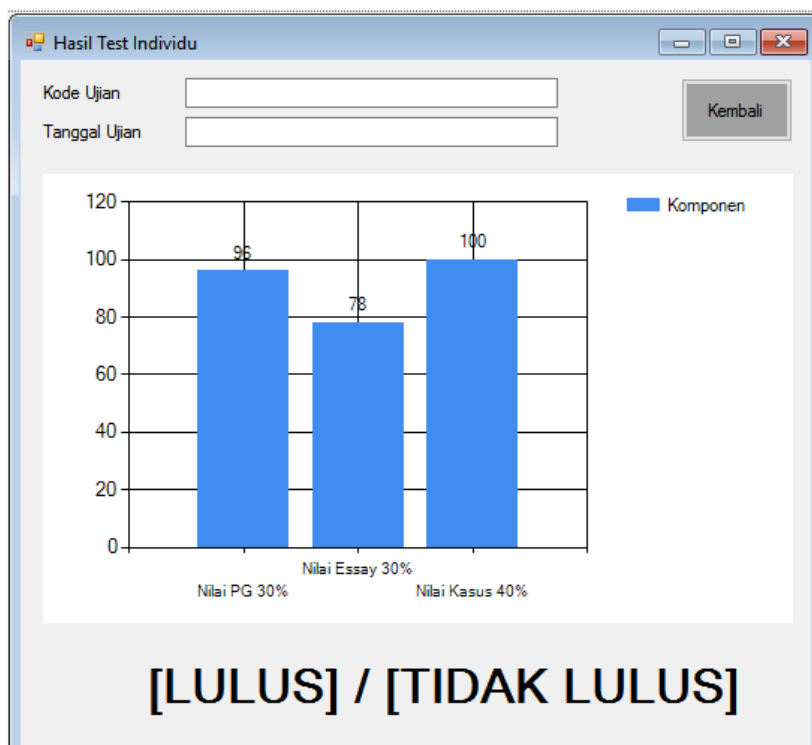
Download Soal Test

Upload Jawaban (max. 3 kali Upload)

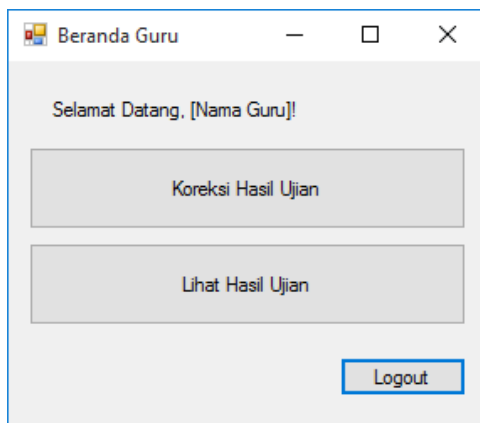
	NamaFile	WaktuUpload
*		

Sisa Waktu Ujian **05:00:00**

09 View Individual Test Result Form



10 Teacher Navigation Form



Beranda Guru

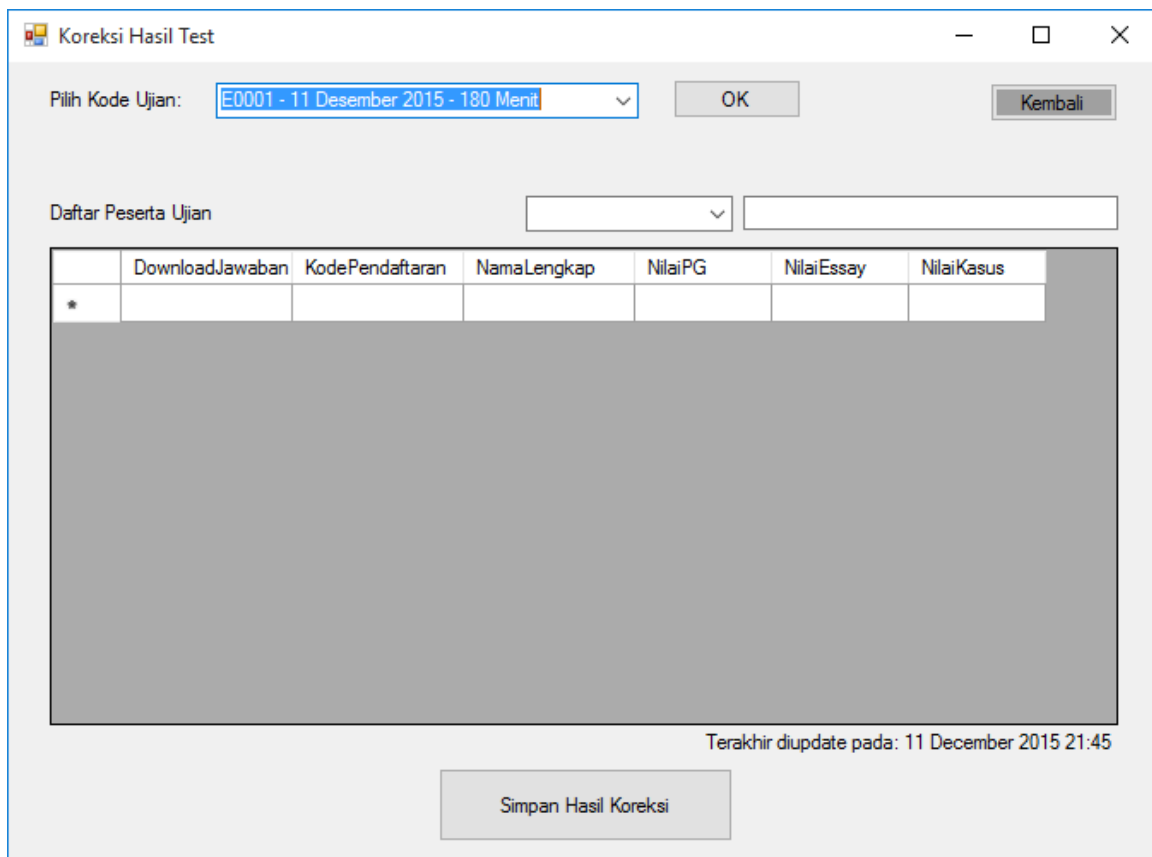
Selamat Datang, [Nama Guru]!

Koreksi Hasil Ujian

Lihat Hasil Ujian

Logout

11 Exam Evaluation Form



Koreksi Hasil Test

Pilih Kode Ujian: E0001 - 11 Desember 2015 - 180 Menit OK Kembali

Daftar Peserta Ujian

	DownloadJawaban	KodePendaftaran	NamaLengkap	NilaiPG	NilaiEssay	NilaiKasus
*						

Terakhir diupdate pada: 11 December 2015 21:45

Simpan Hasil Koreksi

12 View Test Result Form

Rekapitulasi Hasil Test

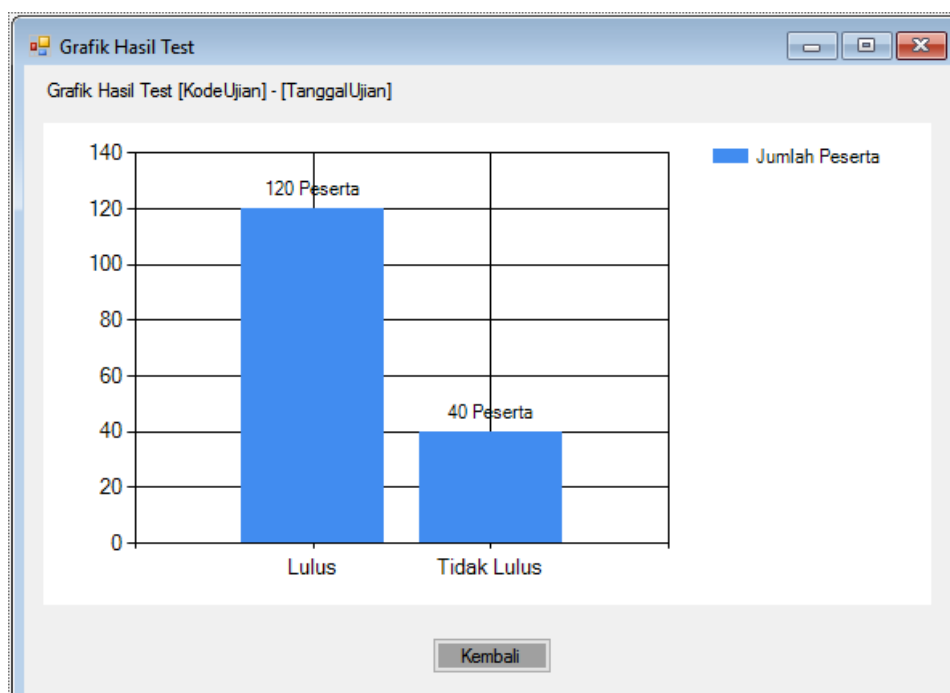
Pilih Kode Ujian

Kembali OK

	KodePendaftaran	NamaLengkap	HasilTest
*			

Lihat Grafik Hasil Test

13 View Test Result Graphic



14 Staff Navigation Form

Beranda Karyawan

Selamat Datang, [Nama Karyawan]!

Logout

Manajemen Master Data

- Karyawan dan Guru
- Calon Siswa
- Jurusan
- Jadwal Ujian
- User Account

Manajemen Ujian

- Alokasi Peserta Ujian
- Koreksi Hasil Ujian
- Laporan Hasil Ujian

15 Employee Management Form

Manajemen Karyawan

Cari Data

Cari

Urutkan Data

Ascending
Descending

Urutkan

	KodeKaryawan	NIP	NamaKaryawan	TanggalLahir	JenisKelamin	No Tele
▶▶						

<

>

Kode Karyawan

K0001

Tambah Data

NIP

0931011992101

Perbarui Data

Nama Karyawan

Ronald Ang

Simpan

Tanggal Lahir

Friday, December 11, 2015

Hapus Data

Jenis Kelamin

☒ Laki-Laki

☐ Perempuan

Kembali

No. Telepon

087809123245

Alamat

Jl. Jalur Sutera Barat, No. 19

16 Study Program Management Form

Manajemen Data Jurusan

	KodeJurusan	NamaJurusan	PenjelasanJurusan	FotoJurusan
▶*				

Kode Jurusan

Nama Jurusan

Penjelasan Jurusan

Foto Jurusan

Browse..

Tambah Data

Perbarui Data

Hapus Data

Simpan

Kembali

17 Applicants / Student Management Form

Manajemen Calon Siswa

Cari Data

Urutkan Data

	KodePendaftaran	NamaLengkap	TanggalLahir	JenisKelamin	NoTelepon	Alamat
▶▶						

Kode Pendaftaran

00001

Perbarui Data

Nama Lengkap

Ferdi Surya

Simpan

Alamat

Jl. Kemanggisan IV. No. 23

Hapus Data

Tanggal Lahir

Friday, December 11, 2015

Kembali

No. Telepon

082113255002

Jenis Kelamin

☒ Laki-Laki
 ☐ Perempuan

Asal Sekolah

SMP Nasional 1

Tgl Pendaftaran

Friday, December 11, 2015

Pilihan Jurusan

Prioritas 1

Rekayasa Piranti Lunak

Prioritas 2

Teknik Komputer dan Jaringan

Prioritas 3

Animasi

Status Test Masuk

[Lulus]

[Tidak Lulus]

[Belum Ujian]

18 Account Management Form

The screenshot shows a window titled "Manajemen User Account". It contains a table with columns "Username", "Password", and "Role". The first row has a "*" in the first column. Below the table are input fields for "Username", "Password", and "Role". The "Role" field has three radio buttons: "Karyawan", "Guru", and "Siswa". At the bottom are three buttons: "Perbarui Data", "Simpan", and "Kembali".

	Username	Password	Role
*			

Username:
Password:
Role: ☐ Karyawan ☐ Guru ☐ Siswa

19 Exam Management Form

The screenshot shows a window titled "Manajemen Ujian". It has search and sort controls at the top. Below is a table with columns "Kode Ujian", "Tanggal Ujian", "Durasi Ujian", "Kapasitas Peserta", and "Soal Ujian". The first row has a "▶*" in the first column. Below the table are input fields for "Kode Ujian", "Tanggal ujian", "Durasi Ujian", and "Kapasitas Peserta". There is also a "Soal Ujian" section with a text input, "Browse...", and "Upload" buttons. On the right side are buttons for "Tambah Data", "Perbarui Data", "Simpan", "Hapus Data", and "Kembali".

Cari Data:
Urutkan Data: ☐ Ascending ☐ Descending

	Kode Ujian	Tanggal Ujian	Durasi Ujian	Kapasitas Peserta	Soal Ujian
▶*					

Kode Ujian:
Tanggal ujian:
Durasi Ujian: Menit Kapasitas Peserta: Orang

Soal Ujian:

20 Student Exam Allocation Form

Alokasi Jadwal Ujian

Jadwal Ujian:

Daftar Siswa

	KodePendaftaran	NamaLengkap
*		

Peserta Ujian Sisa Kursi: 300

	KodePendaftaran	NamaLengkap
*		

>>

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THE DEVELOPMENT OF SMK NUSANTARA INFORMATION SYSTEM

SMK NUSANTARA is one of Indonesia famous vocational school located in Malang, East Java. As the school growth, the Headmaster thinks it is the time to use Information System to support their business process, starting from student registration process. After hiring famous consultant to analyze and design the system, it is the time to develop the system.

Based on the system design given to you, you're required to develop Information system to support the business process of student registration for SMK Nusantara. Ensure your developed solutions can accommodate SMK Nusantara business process properly, following the instruction on the project!

Module 1: Database Creation and Main Form Development

1. Create the Database

Create a database using your MS SQL Server on the local database server following the ERD and Data Dictionary given to you.

2. Insert the Data

Insert the master data given to you on file "*master-data.xlsx*" to the related table on your newly created database.

3. Create Application

Create an application using your preferred platform (C# or Visual Basic.NET).

4. Create "01 Main Form"

Create the main menu of the application as outlined in "**01 Main Form**" in the wireframe.

- Ensure the navigation between forms are created properly referring to the Navigation Diagram given above.

5. Create "02 School Information Form"

Create the main menu of the application as outlined in "**02 School Information Form**" in the wireframe.

- In this form, display the illustrated image of SMK Nasional photo.
- In the section to select Study Program information, ensure this item in the combo box are generated from the database that can be managed from "**16 Study Program Management Form**".

6. Create “03 Detailed Info for Each Study Program Form”

Create the main menu of the application as outlined in “03 Detailed Info for Each Study Program Form” in the wireframe.

- Ensure the title of the form are following the Study Program selected
- Ensure the [Nama Jurusan], [Image], and [Penjelasan mengenai jurusan yang dipilih] field on the form are filled with the selected Study Program data from the database.

7. Create “04 Student Registration Form”

Create the main menu of the application as outlined in “04 Student Registration Form” in the wireframe.

- Validate all fields should be filled
- The priority combo box should be filled from the Study Program data from the database, and the already selected Study Program will not be enlisted in the other combo box, for example:
 - If the registrant choose “Rekayasa Piranti Lunak” in the [Prioritas 1] combo box, then “Rekayasa Piranti Lunak” will not be enlisted again in [Prioritas 2] and [Prioritas 3] combo box.
- Confirmation code should be automatically generated by using random character from a-z, A-Z, number, and symbols. The combination should be 10 characters long, and randomly generated every time this form opened.
- The registrant should enter the confirmation code correctly before submitting their registration application
- Validate the age of registrant must older than 18 years old

8. Create “05 Registration Confirmation Form”

Create the main menu of the application as outlined in “05 Registration Confirmation Form” in the wireframe.

- The data displayed should refer to submitted registrant data, entered in the process from **04 Student Registration Form** before.
- [Nomor Pendaftaran] field should be auto-generated using the formula: **[last registration number] + 1**. For example, if the last registration number are 00009, then the generated number should be 00010. The leading zero number should always be presented!
- The password for the newly created user should be auto-generated by the program using format: **[nomor pendaftaran] + [student birthdate mm/dd/yy]**. For Example, if the registration number are 00010 and the student birthdate are on 31 December 1995, then the password generated should be 00010123195.

9. Create “06 Login Form”

Create the main menu of the application as outlined in “06 Login Form” in the wireframe.

- Ensure the username, password, and role are entered correctly.
- The [error message] should always display the detailed failed login information, with example:
 - “Username not found”. If the entered username are not found in the database

- “Incorrect Password”. If the entered password is not matching the username’s password in the database
- “You do not have permission to enter as this role”. If the role selected is not matching the username’s role in the database.

10. Create “07 Student Navigation Form”

Create the main menu of the application as outlined in “**07 Student Navigation Form**” in the wireframe.

- Ensure the form just can be accessed by student
- Ensure the form can display user Full Name on the field given.
- Ensure the navigation between forms are created properly referring to the Navigation Diagram given above.

11. Create “10 Teacher Navigation Form”

Create the main menu of the application as outlined in “**10 Teacher Navigation Form**” in the wireframe.

- Ensure the form just can be accessed by teacher
- Ensure the form can display user Full Name on the field given.
- Ensure the navigation between forms are created properly referring to the Navigation Diagram given above.

12. Create “14 Staff Navigation Form”

Create the main menu of the application as outlined in “**14 Staff Navigation Form**” in the wireframe.

- Ensure the form just can be accessed by staff
- Ensure the form can display user Full Name on the field given.
- Ensure the navigation between forms are created properly referring to the Navigation Diagram given above.

Module 2: Master Form Development

1. Create “15 Employee Management Form”

Create the main menu of the application as outlined in “**15 Employee Management Form**” in the wireframe.

- Ensure the form can list all Employee data stored inside the database
- Ensure all data are filled before adding or updating the data.
- The Search function will allow the user to choose the fields to be searched for, following the Employee table fields on the database. For example, if the user select “NamaKaryawan” fields, then the search box query should only searching by NamaKaryawan.
- The Sort function will also allow the user to choose the fields to be sorted for, and able to choose the sorting order.
- The Add New, Edit Data, and Delete Data button are functioned correctly.
- As for Save and Cancel button, it only enabled when the user choose to Edit the Data, and should be on the proper state, if the user are editing the data cancel button should cancel the edited data.
- If the user choose to insert new data, then system will automatically generate new 5 characters long employee code, with format “**K**” + **([last EmployeeID] + 1)**. For Example if the last EmployeeID are “K0001” then the employee code generated will be “K0002”
- The password for the newly created employee should be auto-generated by the program using format: **[EmployeeID] + [employee birthdate mm/dd/yy]**.

For Example, if the employee id is K0001 and the employee birthdate are on 31 December 1995, then the password generated should be K0001123195.

- Ensure all general form function are working correctly

2. Create “16 Study Program Management Form”

Create the main menu of the application as outlined in “16 Study Program Management Form” in the wireframe.

- Ensure the form can list all Study Program data stored inside the database
- Ensure all data are filled before adding or updating the data.
- The Add Data, Edit Data, and Delete Data button are functioned correctly.
- As for Save and Cancel button, it only enabled when the user choose to Edit the Data, and should be on the proper state, if the user are editing the data cancel button should cancel the edited data.
- If the user choose to insert new data, then system will automatically generate new 5 characters long study program code, with format “**S**” + ([last StudyProgramID] + 1). For Example if the last Study Program Code are “S0001” then the Study Program Code generated will be “S0002”
- Ensure the program that already have student, cannot be deleted
- Ensure all general form function are working correctly

3. Create “17 Applicant / Student Management Form”

Create the main menu of the application as outlined in “17 Applicant / Student Management Form” in the wireframe.

- Ensure the form can list all Student data stored inside the database
- Ensure all data are filled before adding or updating the data.
- The Search function will allow the user to choose the fields to be searched for, following the Student table fields on the database. For example, if the user select “NamaLengkap” fields, then the search box query should only searching by NamaLengkap.
- The Sort function will also allow the user to choose the fields to be sorted for, and able to choose the sorting order.
- The Edit Data, and Delete Data button are functioned correctly.
- As for Save and Cancel button, it only enabled when the user choose to Edit the Data, and should be on the proper state, if the user are editing the data cancel button should cancel the edited data, if the user are not editing the data, then Cancel button should close the form.
- The priority combo box should be filled from the Study Program data from the database, and the already selected Study Program will not be enlisted in the other combo box, for example:
 - If the registrant choose “Rekayasa Piranti Lunak” in the [Prioritas 1] combo box, then “Rekayasa Piranti Lunak” will not be enlisted again in [Prioritas 2] and [Prioritas 3] combo box.

- [Status Test Masuk] will display the current student progress:
 - If the student is still not taking the test, the form will display [Belum Ujian] as information.
 - If the student Exam Score are already submitted by the teacher, then the status will be automatically updated following the rules below:
 - If the total score are ≥ 65 , then the form will display [Lulus] as information.
 - If the total score are < 65 , then the form will display [Tidak Lulus] as information.
- Ensure all general form function are working correctly

Module 3: Master and Transaction Form Development

1. Create “18 Account Management Form”

Create the main menu of the application as outlined in “**18 Account Management Form**” in the wireframe.

- This form serve to manage all account on the system. All employee that had been registered on the system using **Employee Management Form** will be generated a new username and password. As well as students that have been register using the **Registration Form**
- Every employee can become both Staff and Teacher but not Students, and Students cannot become Staff or Teacher, ensure your database can saved the information correctly and the form developed can validate this requirement.
- For Save and Cancel button, it only enabled when the user choose to Edit the Data, and should be on the proper state, if the user are editing the data cancel button should cancel the edited data, if the user are not editing the data, then Cancel button should close the form.
- Ensure all general form function are working correctly

2. Create “19 Exam Management Form”

Create the main menu of the application as outlined in “19 Exam Management Form” in the wireframe.

- Ensure the form can list all Exam data stored inside the database
- Ensure all data are filled before adding or updating the data.
- The Search function will allow the user to choose the fields to be searched for, following the Exam table fields on the database. For example, if the user select “KodeUjian” fields, then the search box query should only searching by KodeUjian.
- The Sort function will also allow the user to choose the fields to be sorted for, and able to choose the sorting order.
- The Add Data, Edit Data, and Delete Data button are functioned correctly.
- Ensure the date cannot be older than current date.
- Ensure the data cannot be changed if the date already passed.
- As for Save and Cancel button, it only enabled when the user choose to Edit the Data, and should be on the proper state, if the user are editing the data cancel button should cancel the edited data, if the user are not editing the data, then Cancel button should close the form.
- If the user choose to insert new data, then system will automatically generate new 5 characters long Exam ID, with format “E” + ([last ExamID] + 1). For Example if the last ExamID are “E0001” then the Exam ID generated will be “E0002”

- User should be able to upload exam case file to the system, ensure the uploaded exam case are linked to the proper exam schedule. If the user uploaded new exam case, the old files on the system should be overwritten.
- Ensure all general form function are working correctly

3. Create “20 Student Exam Allocation Form”

Create the main menu of the application as outlined in “**20 Student Exam Allocation Form**” in the wireframe.

- This form will be used to allocate student to exam schedule prepared on **Exam Management Form**
- The combo box Exam List should only list all incoming Exam Schedule, using the following format **[ExamID] – [Exam Date]**
- Student List data grid, should only list all students who are not been allocated to any exam schedule only.
- When adding data to Exam Participant data grid, take notes on the exam capacity, if the capacity already maxed, then display information to the user to remove the participant before adding a new one.
- All operation in this form should be automatically saved on the database.

Module 4: Exam and Evaluation Form

1. Create “08 Exam Case Download Form”

Create the main menu of the application as outlined in “**08 Exam Case Download Form**” in the wireframe.

- If the student open this form before the Exam date, then all the links and button in this form will be disabled.
- Student will be able to do the exam within the given exam duration, the timer will be counting down after the students download the exam case.
- Ensure the case downloaded linked to the correct exam schedule.
- When the student click on the upload answer textbox, the program will show the file chooser to browse for the answer file. The student can upload the selected answer file by clicking on the Upload button
- The student will be able to upload their answer in maximum 3 times, within the exam duration. The upload data will be recorded, but only the last uploaded file will be evaluated by the Teacher.
- If the exam duration is ended but the student still not Finalize the answer, the system will automatically finalize student answer.
- Ensure the timer will still counting down even if the program are closed and re-opened by the student.

2. Create “11 Exam Evaluation Form”

Create the main menu of the application as outlined in “11 Exam Evaluation Form” in the wireframe.

- [Pilih Kode Ujian] field should only be filled with Exam schedule that already passed.
- The Search function will allow the user to choose the fields to be searched for between [KodePendaftaran] or [NamaLengkap] field. For example, if the user select “KodePendaftaran” fields, then the search box query should only searching by KodePendaftaran.
- The selected exam participant list will be automatically generated on the data grid. User should be able to directly download the answer by clicking the field [DownloadJawaban] on the data grid, this also applied to input NilaiPG, NilaiEssay, and NilaiKasus.
- When the user submit the evaluation result, the system should automatically calculate the total score for the students using the following formula:

$$\text{Total Score} = (30\% * \text{NilaiPG}) + (30\% * \text{NilaiEssay}) + (40\% * \text{NilaiKasus})$$

- The system also need to update the exam status for all student automatically after submitting the evaluation result with the following criteria:
 - If the total score are ≥ 65 , then the form will display [Lulus] as information.
 - If the total score are < 65 , then the form will display [Tidak Lulus] as information.

- Last updated information should be filled with the last date the user submit the selected exam evaluation result.

3. Create “09 View Individual Test Result Form”

Create the main menu of the application as outlined in “**09 View Individual Test Result Form**” in the wireframe.

- The student can only open this form after the Teacher entry the exam score, if not, show information for the student that his/her score are not available yet, and close the form.
- The form should be able to display student exam schedule and exam score respectively.
- The chart should be able to display detailed student score for [NilaiPG], [NilaiEssay] and [NilaiKasus] as inputted by the Teacher.
- Show the exam status of the student whether he/she passed the exam or not

Module 5: Result Form, ERD and Data Dictionary

1. Create “12 View Test Result Form”

Create the main menu of the application as outlined in “**12 View Test Result Form**” in the wireframe.

- The exam list combo box should only be filled with exam schedule that already completely evaluated by the teacher. If there is missing student which not had been evaluated on the exam schedule, then the system will not show the schedule.

2. Create “13 View Test Result Graphic”

Create the main menu of the application as outlined in “**13 View Test Result Graphic**” in the wireframe.

- Develop chart to display number of student who pass the exam and not on the selected exam schedule.
- Ensure the information for selected exam date are displayed using correct format.

3. Create “ERD and Data Dictionary”

SMK Nusantara is going to continue the development of their Information System to support another business process of the school. The next phases are going to develop the system for Allocating Student Class.

Usually, all students are allocated to their class by considering the Entrance Exam Score to group them to these following classes:

1. Global Class, which all lecturing activity will be conducted in English, this class is suited for all students who got TOEFL score higher than 550 and having Entrance Exam Score higher than 90.
2. High Achiever Class, suited for all students who got Entrance Exam Score higher than 80.
3. Normal Class, suited for all students who do not meet the criteria for Global and High Achiever Class.

Another thing to analyze, all class will have one home-class teacher; each teacher can become home-class teacher in one class only.

4. Giving the simple constraint above, please design the ERD and Data Dictionary for new table and new field to support Student Class Allocation in SMK Nusantara. Please refer to existing ERD and Data Dictionary to develop your design solutions.